

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

Human Resource Management (HRM) is the systematic approach to managing employees and organizational workforce activities in an efficient and structured manner. In today's fast-paced organizational environment, managing employee information, pre-joining onboarding, daily attendance, leave requests, payroll-related information, compliance requirements, and employee exit activities manually or through disconnected systems can create significant operational challenges.

The Employee Onboarding & HR Management System is designed as a centralized digital solution that streamlines and manages major activities throughout the employee lifecycle. The system is developed using a modern decoupled architecture consisting of a Spring Boot backend REST API and a React-based responsive frontend application. It provides centralized employee data management, employee onboarding, attendance management, leave management, payroll management, compliance management, HR analytics, and employee exit management.

Traditional human resource operations often rely on physical documents, spreadsheets, emails, and manual data entry. As the number of employees increases, maintaining employee records, verifying onboarding documents, tracking attendance, managing leave balances, processing payroll information, and monitoring employee status can become time-consuming and error-prone. Manual processes can also result in duplicate records, delayed approvals, inconsistent information, and difficulty in accessing updated employee data.

The proposed system establishes a centralized source of employee information and uses secure authentication and Role-Based Access Control (RBAC) to provide appropriate access to different categories of users. The system supports roles such as SUPER_ADMIN, HR_MANAGER, DEPT_MANAGER, and EMPLOYEE, with each role having specific permissions. By automating and centralizing HR activities, the platform reduces manual effort, improves data consistency, enhances transparency, and provides a more efficient approach to employee lifecycle management.

1.2 COMPONENTS OF SYSTEM

The Employee Onboarding & HR Management System consists of several integrated operational components, each designed to manage a specific stage of workforce administration.

1. Employee Onboarding Portal

The Employee Onboarding component manages the process of adding and onboarding new employees into the organization. HR personnel can maintain employee details, onboarding information, required documents, offer letter information, and onboarding status. The component provides a structured workflow for tracking the progress of new employees before they become active members of the organization.

2. Employee Profile & Directory Management

The Employee Management component maintains centralized employee records. Authorized HR personnel can create, view, update, and manage employee information such as employee ID, name, contact details, department, designation, joining date, employment status, and other relevant organizational information. The centralized directory makes employee information easier to access and maintain.

3. Attendance Management

The Attendance Management component allows the organization to maintain employee attendance records. HR managers and authorized users can monitor attendance information, while employees can view their attendance details. The system also supports attendance regularization, allowing employees to submit requests for correcting attendance-related issues.

4. Leave Management

The Leave Management component enables employees to apply for leave and monitor the status of their requests. Authorized managers can review, approve, or reject leave applications based on their assigned permissions. The system maintains leave-related information and provides employees with better visibility of their leave status.

5. Payroll Management

The Payroll Management component provides centralized management of employee payroll information. HR personnel can maintain salary-related details and access payroll records. The module helps organize payroll information and reduces the need for maintaining separate manual records.

6. Compliance Management

The Compliance Management component helps HR personnel maintain employee-related compliance information and required records. It provides a centralized approach for monitoring relevant employee documents and compliance activities and helps HR users keep track of important requirements.

7. HR Analytics & Workforce Management

The HR Analytics component provides useful workforce information such as employee headcount and department-level employee information. The analytics dashboard helps HR personnel understand workforce data and supports data-driven administrative decision-making.

8. Employee Exit Management

The Exit Management component supports employee resignation and exit-related activities. It helps HR personnel maintain resignation information, employee exit status, clearance-related details, and other activities associated with the employee's departure from the organization.

9. Authentication & Role-Based Access Control

The authentication component provides secure login and authorization for system users. JWT-based authentication is used to protect application resources, while Role-Based Access Control ensures that users can access only the modules and operations permitted for their assigned roles.

10. Database Management

The system uses a relational database to centrally store employee, attendance, leave, payroll, onboarding, compliance, and other HR-related information. Database integration ensures structured storage, retrieval, and management of organizational data.

1.3 ADVANCED TECHNOLOGIES

The platform uses modern software technologies and development frameworks to provide security, maintainability, and efficient communication between the frontend and backend.

- **JWT-Based Authentication with RBAC**

JSON Web Token (JWT) is used for user authentication and secure access to protected resources. After successful login, an authenticated user receives a token that is used when accessing protected APIs. Role-Based Access Control provides different permissions for roles such as SUPER_ADMIN, HR_MANAGER, DEPT_MANAGER, and EMPLOYEE.

- **Spring Boot REST API**

Spring Boot is used to develop the backend application and RESTful APIs. The backend handles business logic, authentication, authorization, employee management, attendance, leave, payroll, compliance, and analytics-related operations.

- **React-Based Frontend**

React is used to develop the responsive frontend application. Its component-based architecture supports the development of reusable user interface components and role-specific dashboards for employees, managers, and HR personnel.

- **Spring Security**

Spring Security is used to implement authentication and authorization mechanisms. Passwords are securely stored using BCrypt hashing, while protected API endpoints are controlled according to user roles and permissions.

- **MySQL Database**

MySQL is used as the primary relational database for storing and managing HRMS information. The database maintains structured records related to employees, attendance, leave, payroll, onboarding, compliance, and other system modules.

- **JPA and Hibernate**

Java Persistence API (JPA) and Hibernate are used for object-relational mapping between the Java application and the relational database. They simplify database operations and allow application entities to be mapped to database tables.

- **Swagger/OpenAPI**

Swagger/OpenAPI is used to document and test the REST APIs. It provides an interactive interface through which developers can understand available endpoints, request parameters, responses, and authentication requirements.

1.4 GLOBAL PERSPECTIVES & INDUSTRY STANDARDS

Global corporate trends emphasize digital transformation, centralized workforce management, employee self-service, data security, and data-driven HR operations. The proposed Employee Onboarding & HR Management System follows commonly adopted software engineering and security practices to provide a structured and secure HR management environment.

1. Software Requirements Specification

The system documentation follows an organized Software Requirements Specification (SRS) structure covering functional requirements, non-functional requirements, external interfaces, system features, constraints, assumptions, and dependencies.

2. Role-Based Access Control

Role-Based Access Control is used to restrict system functionality according to the responsibilities of each user. This approach helps prevent unauthorized access to sensitive employee and organizational information.

3. Secure Password Management

User passwords are stored using BCrypt password hashing rather than storing passwords in plain text. This provides an additional layer of protection for user authentication information.

4. RESTful Architecture

The backend follows a RESTful API architecture to enable communication between the React frontend and Spring Boot backend. This separation improves maintainability and allows frontend and backend components to be developed independently.

5. Data Privacy and Security

Employee information may contain sensitive personal and employment-related data. Therefore, the system uses authentication, authorization, controlled database access, and secure password storage to protect information from unauthorized access.

6. OWASP Security Practices

The system follows relevant secure development practices such as input validation, authentication, authorization, secure password hashing, controlled API access, and protection against common web application security risks.

7. Digital Employee Lifecycle Management

Modern HR platforms increasingly focus on managing employees throughout their lifecycle. The proposed system integrates onboarding, employee management, attendance, leave, payroll, compliance, analytics, and exit management into a centralized platform.

Overall, the Employee Onboarding & HR Management System provides a structured digital approach to HR operations by reducing dependency on manual processes, improving information accessibility, strengthening access control, and supporting efficient management of the employee lifecycle.

CHAPTER 2

SYSTEM ANALYSIS

2.1 EXISTING SYSTEM

In many small-to-medium enterprise (SME) organizations and conventional corporate environments, human resource processes are managed through manual or semi-automated methods. Onboarding new employees may involve physical document collection, email communication, manual verification, and spreadsheet-based record keeping. Employee attendance may be maintained through separate biometric systems, manual registers, or spreadsheets, requiring HR personnel to collect and process attendance information periodically. Payroll-related information is also often maintained using spreadsheets and separate applications, which can increase the possibility of calculation and data-entry errors.

The existing approach frequently relies on decentralized communication for HR activities. Leave applications may be submitted through paper forms, emails, or separate communication channels, while department managers may maintain team attendance and leave information independently. Employee records can therefore become distributed across spreadsheets, documents, emails, and different systems, making it difficult for HR personnel to maintain a consistent and updated source of information.

When employee-related information is maintained across multiple systems, HR personnel may spend considerable time collecting, verifying, and updating information. Generating reports, monitoring employee status, tracking attendance and leave, and maintaining payroll-related records can also become time-consuming. These limitations demonstrate the need for a centralized system that integrates major HR activities into a single platform.

2.1.1 DRAWBACKS OF EXISTING SYSTEM

The conventional HR management approach can result in several technical, operational, and administrative challenges.

Quality of Service Issues

The employee onboarding experience can be inconsistent when processes depend heavily on manual documentation and communication. Delays in document verification,

leave approvals, attendance corrections, and other HR activities can affect the efficiency of HR operations and employee experience.

Limited Functionality and Integration Gaps

Separate systems for employee management, attendance, leave, payroll, and other HR activities can create integration gaps. Information may need to be entered or transferred manually between systems, resulting in duplicated work and difficulty maintaining consistent records.

Risk of Calculation and Data Errors

Manual data entry and spreadsheet-based calculations can result in incorrect employee information, attendance records, leave balances, or payroll-related calculations. Errors may require additional verification and correction by HR personnel.

Security and Privacy Vulnerabilities

HR systems contain sensitive employee information, including personal details, employment information, attendance records, and payroll-related data. Storing such information in spreadsheets or unsecured files can increase the risk of unauthorized access or accidental disclosure.

Operational Bottlenecks and Administrative Effort

Manual HR processes require significant time for data entry, verification, approval, reporting, and record maintenance. As the number of employees increases, these activities become more difficult to manage efficiently.

2.2 PROBLEM DEFINITION

The existing HR management approach creates difficulties in maintaining centralized employee information, automating onboarding activities, managing attendance and leave, organizing payroll information, maintaining compliance records, and generating workforce insights.

The problem can therefore be defined as follows:

Organizations require a centralized, secure, and integrated web-based HR management system that can simplify employee onboarding, employee record management, attendance,

leave, payroll, compliance, analytics, and employee exit activities while providing controlled access to information based on user roles.

The system should reduce dependency on manual processes, improve data consistency, simplify HR operations, and provide employees and authorized HR personnel with convenient access to relevant information.

2.3 PROPOSED SYSTEM

The proposed Employee Onboarding & HR Management System overcomes the limitations of conventional HR processes by providing a centralized web-based platform for managing major employee lifecycle activities. The system uses a Spring Boot backend REST API and a React-based frontend application to provide communication between the user interface and backend services.

The system supports different user roles, including SUPER_ADMIN, HR_MANAGER, DEPT_MANAGER, and EMPLOYEE. Each role is provided with appropriate access permissions based on its responsibilities. Employees can access relevant personal and HR information, while HR managers and authorized administrators can manage employee records and HR activities.

The proposed system integrates employee onboarding, employee management, attendance management, leave management, payroll management, compliance management, HR analytics, and employee exit management into a unified platform. This centralized approach reduces the need for maintaining separate records and improves the accessibility and consistency of HR information.

Authentication and authorization are implemented using JWT-based authentication and Role-Based Access Control (RBAC). Passwords are securely stored using BCrypt hashing, and protected APIs are accessible only to authorized users. The system also provides API documentation and testing support through Swagger/OpenAPI.

2.3.1 ADVANTAGES OF PROPOSED SYSTEM

Key technical and operational benefits of the proposed solution include:

- **Centralized Employee Management**

All major employee-related information is maintained in a centralized database. This reduces dependency on separate spreadsheets and documents and makes employee information easier to manage.

- **Improved Employee Onboarding**

The system provides a structured onboarding process for adding and managing new employees. HR personnel can maintain onboarding information and monitor the employee's onboarding status.

- **Integrated HR Modules**

Employee management, attendance, leave, payroll, compliance, analytics, and exit management are provided within a single system, reducing the need to use multiple disconnected applications.

- **Reduced Manual Errors**

Centralized data management and automated system workflows reduce repetitive data entry and help minimize errors associated with manual HR processes.

- **Enhanced Security and Access Control**

JWT authentication, Role-Based Access Control, Spring Security, and BCrypt password hashing help protect the system and restrict access to authorized users.

- **Better HR Decision Making**

The HR analytics module provides workforce-related information such as employee headcount and other available HR metrics, helping HR personnel monitor organizational information more effectively.

- **Improved Accessibility**

The web-based application allows authorized users to access HR functions through a centralized interface, making routine HR activities more convenient.

- **Time-Saving and Cost-Effective**

By reducing paperwork, repetitive data entry, and manual record maintenance, the system helps reduce administrative effort and improves the efficiency of HR operations.

CHAPTER 3

SYSTEM REQUIREMENTS

The hardware and software requirements and the platform description of the system are explained under Sections 3.1, 3.2 and 3.3 respectively.

3.1 HARDWARE REQUIREMENTS

The hardware configuration used for developing and testing the Employee Onboarding & HR Management System is given below.

1. Processor Type : 12th Gen Intel® Core™ i5-12450H
2. RAM : 16 GB RAM
3. Hard Disk : 951.6 GB

3.2 SOFTWARE REQUIREMENTS

- a. Operating System : Windows 11
- b. Front End : React.js
- c. Back End : Spring Boot
- d. Coding Language : Java, ReactJs
- e. Database : MySQL

3.3 SOFTWARE DESCRIPTION



Fig. 3.1. Visual Studio Code

Visual Studio Code, commonly known as VS Code, is a lightweight source-code editor developed by Microsoft. It supports multiple programming languages and development frameworks and can be used on Windows, Linux, and macOS. It provides features such as syntax highlighting, intelligent code completion, debugging support, integrated terminal, Git integration, and extension support. In this project, Visual Studio Code is used for developing and managing the React.js frontend application.

3.3.1 FRONTEND

React.js

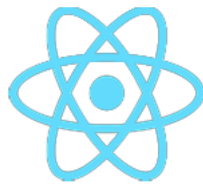


Fig. 3.2 React Logo

React.js is a JavaScript library used for building interactive and responsive user interfaces for web applications. React follows a component-based architecture in which the user interface is divided into reusable components. This approach makes the application easier to develop, maintain, and modify.

In the Employee Onboarding & HR Management System, React.js is used to develop different screens such as the login page, employee dashboard, HR dashboard, employee management, attendance management, leave management, payroll management, compliance management, and other system interfaces.

React uses a Virtual DOM to efficiently update the user interface when application data changes. Instead of updating the complete web page, React identifies the required changes and updates the corresponding elements. This helps provide a responsive user experience.

React also uses JSX, which allows developers to write HTML-like structures within JavaScript code. JSX makes React components easier to organize and understand.

Features of React.js

1. Declarative

React uses a declarative programming approach that allows developers to describe how the user interface should appear based on the current application state.

2. Component-Based Architecture

React applications are divided into reusable and independent components. Components can be created for different sections of the HRMS application and reused wherever required.

3. Virtual DOM

React maintains a Virtual DOM and updates only the required parts of the user interface when application data changes. This improves the efficiency of UI updates.

4. JSX

JSX allows developers to write HTML-like syntax inside JavaScript. It makes the structure of React components easier to read and maintain.

5. One-Way Data Flow

React follows a unidirectional data flow in which data generally moves from parent components to child components. This provides a predictable structure for managing application data.

6. Reusable Components

React allows commonly used interface elements to be developed as reusable components. This reduces code duplication and simplifies application maintenance.

7. Responsive User Interface

React can be used to create responsive interfaces that allow users to access HRMS functions through different screen sizes and devices.

3.3.2 BACKEND

Java

Java is a high-level, object-oriented programming language widely used for developing enterprise and web applications. Java provides platform independence, strong type checking, automatic memory management, and a comprehensive collection of libraries and APIs.

In the Employee Onboarding & HR Management System, Java is used as the primary backend programming language. It is used to implement business logic, REST APIs, authentication, authorization, database operations, and other backend functionalities.

Spring Boot



Fig. 3.3 SpringBoot Logo

Spring Boot is a Java-based framework used for developing web applications and RESTful APIs. It simplifies application development by providing auto-configuration, dependency management, embedded server support, and integration with other Spring technologies.

In this project, Spring Boot is used to develop the backend REST API that communicates with the React.js frontend and MySQL database.

Extensions and Tools Used for Backend Development

1. Extension Pack for Java

The Extension Pack for Java provides essential Java development features such as code completion, debugging, project management, and Java language support.

2. Spring Boot Support

Spring Boot development tools provide support for creating, managing, and developing Spring Boot applications.

3. Maven

Maven is used as the build and dependency management tool for the Spring Boot application. It manages project dependencies and helps compile, package, and run the backend application.

Dependencies Used to Build the Project

1. Spring Web

Spring Web provides the required components for developing RESTful web services. It is used to create controllers, handle HTTP requests, and return responses between the frontend and backend.

2. Spring Data JPA

Spring Data JPA simplifies database access by providing repository-based data operations. It allows Java entity classes to be mapped to database tables and supports operations such as creating, retrieving, updating, and deleting records.

3. MySQL Driver

The MySQL Driver allows the Spring Boot application to communicate with the MySQL relational database. It provides the required connectivity between the backend application and the database.

4. Spring Security

Spring Security is used to implement authentication and authorization in the application. It protects restricted resources and ensures that only authorized users can access specific system functionalities.

5. JWT

JSON Web Token (JWT) is used for token-based authentication. After successful login, the system generates a JWT token that is used to access protected APIs.

6. BCrypt

BCrypt is used for securely hashing user passwords before storing them in the database. It prevents passwords from being stored as plain text.

7. Lombok

Lombok is a Java library that reduces repetitive boilerplate code. It can automatically generate commonly used methods such as getters, setters, constructors, and other methods using annotations.

8. Spring Boot DevTools

Spring Boot DevTools provides development-time features that improve the development process, such as automatic application restart and faster development feedback.

3.3.3 DATABASE

MySQL



Fig. 3.2. MySQL Database

MySQL is a relational database management system used to store and manage structured application data. It uses SQL for creating, retrieving, updating, and deleting data.

In the Employee Onboarding & HR Management System, MySQL is used to store employee information and records related to different HR modules. The database contains information associated with employee management, onboarding, attendance, leave, payroll, compliance, and other system operations.

The Spring Boot backend communicates with the MySQL database using Spring Data JPA and Hibernate. This provides a structured method for storing and retrieving information from the database.

The key features of MySQL used in the project are:

1. Relational Data Management :

Stores project information in structured tables with relationships established using primary keys and foreign keys.

2. Data Integrity:

Uses constraints such as PRIMARY KEY, FOREIGN KEY, NOT NULL, UNIQUE, and CHECK to maintain accurate and consistent data.

CHAPTER 4

SYSTEM DESIGN

4.1 MODULE DESCRIPTION

The Employee Onboarding & HR Management System consists of several integrated modules designed to manage the complete employee lifecycle. The major modules of the system include Employee Onboarding, Employee Management, Attendance Management, Leave Management, Payroll Management, Compliance Management, and HR Analytics & Exit Management. Each module performs specific HR-related functions and works together to provide a centralized and efficient platform for managing employee information and organizational activities

4.1.1 USER & SECURITY MANAGEMENT

The User & Security Management module handles authentication, role-based access control (RBAC), and security management. It allows users (Employee, Department Manager, HR Admin) to log in securely, update their account passwords, and maintain active authentication tokens.

Table 4.1.1: User & Security Management

Field	Description
User ID	Unique identifier (UUID) for each user account in the system
Email	User's official work email address, used for login identification
Password	BCrypt encrypted password hash for secure authentication
Role	System access tier (EMPLOYEE, DEPARTMENT_MANAGER, HR_MANAGER, SUPER_ADMIN)
Active Status	Boolean indicator (true / false) representing user login eligibility

4.1.2 PRE-JOINING ONBOARDING MANAGEMENT

The Pre-Joining Onboarding module enables HR administrators to create digital offer letters, track candidate e-signatures, monitor background verification (BGV), and perform automated Day 1 conversion of candidates into active employee master records.

Table 4.2: Pre-Joining Onboarding Management

Field	Description
Onboarding ID	Unique identifier for each candidate onboarding tracking record
Candidate Name	Full name of the recruited candidate
Candidate Email	Personal or provisional work email of the candidate
Offered CTC	Total annual Cost to Company (CTC) offered in Rupees
e-Sign Status	Offer letter acceptance indicator (Completed / Pending)
BGV Status	Background verification result (PENDING / VERIFIED_PASSED)
Status	Pipeline status (OFFER_GENERATED, OFFER_ACCEPTED, BGV_COMPLETED, ONBOARDED)

4.1.3 EMPLOYEE MASTER DIRECTORY MANAGEMENT

The Employee Master Directory module maintains complete organizational records for all active, probation, and notice period staff. It tracks designation, department hierarchy, manager mappings, gross salary structures, and bank account parameters.

Table 4.3: Employee Master Directory Management

Field	Description
Employee ID	Unique primary key (UUID) assigned to each employee record
Employee Code	Human-readable organizational ID code (e.g., EMP1001, EMP1002)
Department	Organizational unit assignment (e.g., <i>Engineering, HR, Finance, Marketing</i>)
Designation	Job title assigned to the employee
Gross Salary	Total monthly gross earnings in Rupees
Manager ID	Reference to the assigned reporting Department Manager
Status	Employment lifecycle state (ACTIVE, PROBATION, NOTICE, EXITED, TERMINATED)

4.1.4 ATTENDANCE & LEAVE MANAGEMENT

The Attendance & Leave Management module enables employees to punch daily in/out times, apply for leave quotas (Casual, Sick, Earned, Maternity/Paternity), and allows department managers to approve or reject leave requests and regularization claims.

Table 4.4: Attendance & Leave Management

Field	Description
Record ID	Unique identifier for attendance punch or leave application entries
Employee ID	Reference to the employee submitting the attendance or leave entry
Work Date / Range	Date of punch log or start/end dates of leave application
In/Out Punch Time	Timestamps recorded for daily shift check-in and check-out
Leave Type	Categorization of leave requested (CL, SL, EL, COMP_OFF, MATERNITY)
Status	Approval lifecycle status (PENDING, APPROVED, REJECTED, PRESENT, LATE)

4.1.5 PAYROLL & FNF SETTLEMENT MANAGEMENT

The Payroll & FnF Settlement module handles automated salary component calculations (Basic, HRA, Special Allowance), monthly payslip generation, multi-department No-Dues clearances (Manager, IT, Admin, Finance), and Full & Final (FnF) exit settlements.

Table 4.5: Payroll & FnF Settlement Management

Field	Description
Exit / Payroll ID	Unique identifier for monthly payroll run or exit settlement record
Employee ID	Reference to the employee undergoing payroll processing or exit settlement
Department Clearances	Four-way no-dues sign-offs (Manager, IT, Admin, Finance clearances)
EL Encashment	Calculated encashment payout based on unused Earned Leave balance

FnF Settlement Amount Computed net payout amount settled upon complete clearance sign-off

FnF Status Overall settlement status (PENDING, SETTLED, EXITED)

4.2 USE CASE DIAGRAM

A **Use Case Diagram** in UML represents the interactions between users and the Employee Onboarding & HR Management System. It shows the main actors and the functions they can perform.

Actors:

1. **Employee:** Login, manage profile, mark attendance, apply leave, view payslip, and submit resignation.
2. **Department Manager:** View attendance, approve leave, provide clearance, and manage appraisals.
3. **HR Admin / Super Admin:** Manage employees, onboarding, BGV, payroll, clearances, and FnF settlement.

UML USE CASE DIAGRAM: EMPLOYEE ONBOARDING AND HR MANAGEMENT SYSTEM

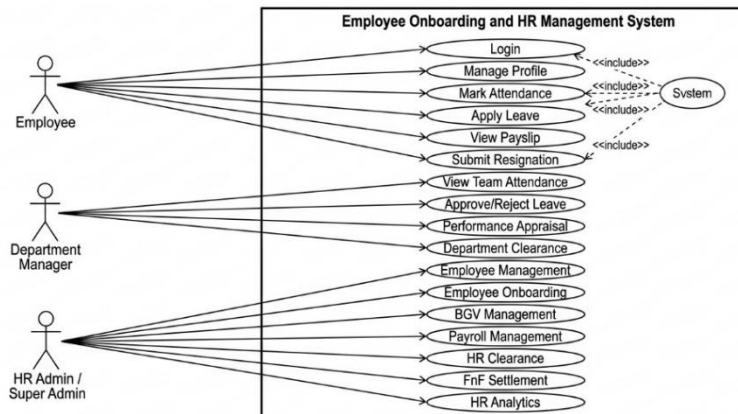


Fig.4.1. Use Case Diagram

4.3 SEQUENCE DIAGRAM

A **Sequence Diagram** represents the chronological order of interactions between the Employee, React Client, REST API Controller, Exit Service, and MySQL Database during the resignation, clearance, and Full & Final (FnF) settlement process.

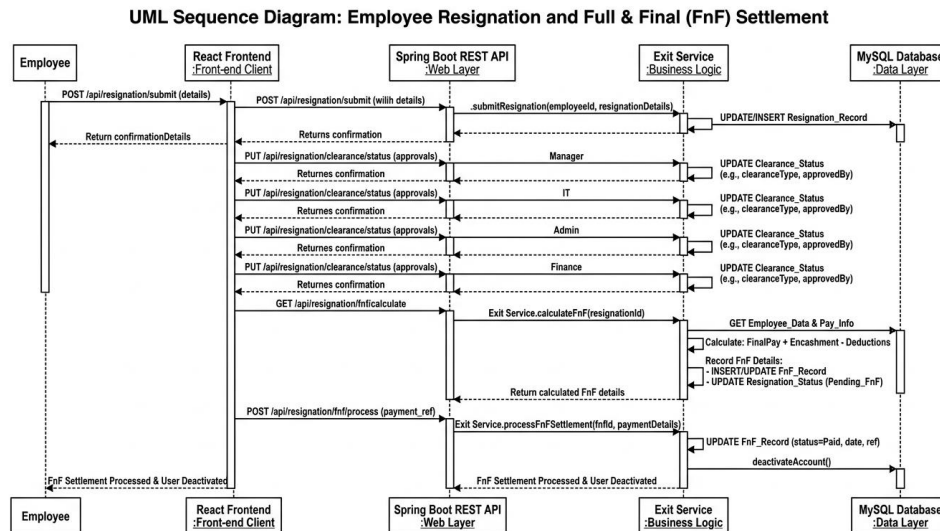


Fig. 4.2. Sequence Diagram

4.4 DATA FLOW DIAGRAM

A Data Flow Diagram (DFD) is a visual representation of how data moves through the Employee Onboarding & HR Management System. It shows the interaction between users, system processes, databases, and external entities. It also illustrates the flow of employee details, attendance records, leave requests, payroll information, onboarding documents, and exit information

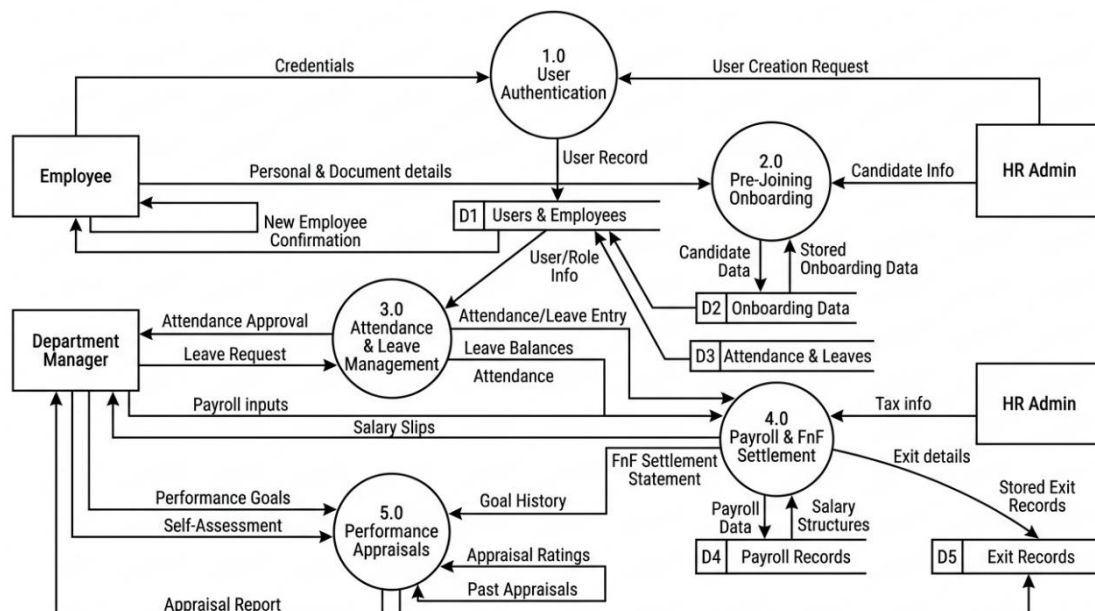


Fig. 4.3. Dataflow Diagram

CHAPTER 5

TESTING

5.1 UNIT TESTING

Unit testing forms the foundational layer of the testing process for the Employee Onboarding & HR Management System. It involves testing individual software components independently to identify and fix errors at an early stage. The testing focuses on Java classes, service methods, controllers, and database-related operations to ensure that each component performs its intended function correctly.

5.2 INTEGRATION TESTING

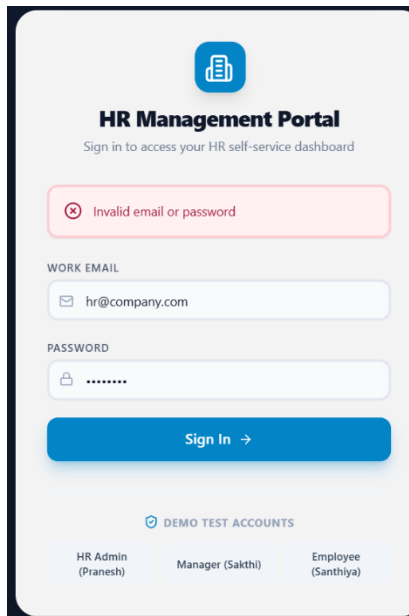
Integration testing focuses on verifying the interaction between different modules of the Employee Onboarding & HR Management System. It ensures that the React frontend, REST API, backend services, and MySQL database work together correctly. API endpoints for employee onboarding, attendance, leave, payroll, compliance, and other HR operations are tested to ensure proper data flow and communication between system components.

5.3 SECURITY AND AUTHENTICATION

Security and authentication are important components of the Employee Onboarding & HR Management System. The system uses JSON Web Tokens (JWT) for secure user authentication and authorization. User credentials are securely stored using password encryption. After successful login, a JWT is generated and provided to the authenticated user.

The token is included in subsequent API requests using the Bearer Token format. The backend validates the token before allowing access to protected resources. Role-Based Access Control (RBAC) is used to restrict access according to user roles such as SUPER_ADMIN, HR_MANAGER, DEPT_MANAGER, and EMPLOYEE. JWT expiration further improves security by requiring users to authenticate again after the token expires.

5.4.1 TEST CASE I



Test ScenarioI: Login with invalid or missing credentials.

EXPECTED OUTPUT: The system should reject the login attempt and display an appropriate error message indicating that the username or password is incorrect.

ACTUAL OUTPUT: The login attempt is rejected successfully, and the system displays Invalid email or password.

5.4.2 TEST CASE II

✓ Candidate successfully onboarded as Employee EMP3683!

Search candidates...

Candidate ¹	Email	Role	Offered CTC	e-Sign	BGV Status	Status	Action
Dhanesh	dhanesh@company.com	Fresher (Engineering)	₹800,000	Mark Signed	Mark BGV	OFFER GENERATED	Day 1 Onboard
SK	praneshthirumalaivrt@gmail.com	IT Cousantant (Marketing)	₹5,000,000	Mark Signed	Mark BGV	ONBOARDED	Onboarded
vbm	praneshthirumalaivrt@gmail.com	IT (Finance)	₹800,000	Mark Signed	Mark BGV	OFFER GENERATED	Day 1 Onboard
Dhanesh	Dhanesh@gmail.com	Junior Enginner (Marketing)	₹500,000	Mark Signed	Mark BGV	ONBOARDED	Onboarded
Murgavel	murga@gmail	Enginner (Engineering)	₹500,000	Mark Signed	Mark BGV	ONBOARDED	Onboarded
John Doe	johndoe@example.com	Frontend Developer (Engineering)	₹undefined	Mark Signed	Mark BGV	ONBOARDED	Onboarded
John Doe	johndoe@example.com	Frontend Developer (Engineering)	₹undefined	Mark Signed	Mark BGV	OFFER ACCEPTED	Day 1 Onboard
Raj	raj@company.com	Freshers (Engineering)	₹800,000	Mark Signed	Passed	BGV COMPLETED	Day 1 Onboard
Alex	Alex@gmail.com	Enginner (Human Resources)	₹800,000	Mark Signed	Passed	BGV COMPLETED	Day 1 Onboard
Iyal	iyal@company.com	Fresher (Engineering)	₹800,000	Mark Signed	Mark BGV	OFFER GENERATED	Day 1 Onboard

Showing 1 to 10 of 10 entries

< Page 1 of 1 >

Test ScenarioII: Successful employee onboarding.

EXPECTED OUTPUT: After successfully submitting the candidate details, the system should create the employee record and display a confirmation message indicating that the candidate has been successfully onboarded.

ACTUAL OUTPUT: The candidate was successfully onboarded as Employee EMP3683, and the system displayed the confirmation message Candidate successfully onboarded as Employee EMP3683.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

The Employee Onboarding & HR Management System successfully addresses the challenges of managing employee information and HR activities through a centralized digital platform. Traditional HR processes often depend on manual records, spreadsheets, emails, and separate systems, which can result in data duplication, delays, and errors. The proposed system provides an integrated solution for managing the complete employee lifecycle, from pre-joining onboarding and document verification to attendance, leave, payroll, performance appraisal, and employee exit management.

The system is developed using modern technologies such as Spring Boot, React.js, MySQL, Spring Security, and JWT authentication. It provides role-based access for Super Admin, HR Manager, Department Manager, and Employee, ensuring that users can access only the functions relevant to their roles.

The major benefits of the system include:

- Centralized management of employee information and documents.
- Automated employee onboarding and offer letter management.
- Efficient attendance and leave management.
- Automated payroll processing and statutory deduction calculation.
- Secure authentication and role-based access control.
- Performance appraisal and employee development management.
- HR analytics for better workforce monitoring and decision-making.
- Streamlined resignation, clearance, and Full & Final (FnF) settlement processes.

Overall, the Employee Onboarding & HR Management System improves HR operational efficiency, reduces manual work, enhances data security, and provides a unified platform for managing employee-related activities throughout the organization.

6.2 FUTURE WORK

The current system provides a strong foundation for digital HR management. The following enhancements can be implemented in future versions:

- **Mobile Application:** Develop Android and iOS applications for employees to access attendance, leave, payslips, notifications, and other HR services remotely.
- **AI-Based HR Analytics:** Integrate Artificial Intelligence and Machine Learning to analyze employee data, predict attrition trends, identify workforce patterns, and support HR decision-making.
- **Biometric Attendance Integration:** Integrate biometric devices for automatic attendance synchronization and real-time employee attendance tracking.
- **Advanced Payroll Integration:** Enhance integration with banking and payment services for automated salary disbursement and payroll processing.
- **Chatbot for HR Support:** Introduce an AI-powered chatbot to answer common employee queries related to leave, attendance, payroll, policies, and other HR services.
- **Advanced Notifications:** Implement automated email, SMS, and mobile notifications for onboarding, leave approvals, attendance alerts, payroll, and exit processes.
- **Cloud Deployment:** Deploy the system on cloud platforms to improve scalability, availability, backup, and accessibility.
- **Enhanced Security:** Add Multi-Factor Authentication (MFA), biometric verification, and advanced audit logging to further strengthen system security.

CHAPTER 7

APPENDIX I SOURCE CODE

HR Dashboard

```
employee-onboarding-system > frontend > src > pages > HRDashboard.jsx > HRDashboard
1 import React, { useEffect, useState } from 'react';
2 import StatCard from '../components/common/StatCard';
3 import { analyticsApi } from '../api/analyticsApi';
4 import { Users, UserPlus, Calendar, DollarSign, Clock, Logout, CheckSquare, ShieldCheck } from 'lucide-react';
5 import { ResponsiveContainer, AreaChart, Area, XAxis, YAxis, Tooltip, PieChart, Pie, Cell } from 'recharts';
6
7 export default function HRDashboard() {
8   const [data, setData] = useState(null);
9
10  useEffect(() => {
11    loadAnalytics();
12  }, []);
13
14  const loadAnalytics = async () => {
15    try {
16      const res = await analyticsApi.getAnalytics();
17      setData(res.data);
18    } catch (e) {
19      console.error(e);
20    }
21  };
22
23  const COLORS = ['#00286c', '#a10898', '#f59e0b', '#8b5cf6', '#ef4444'];
24  const deptData = data?.headcountByDepartment
25    ? Object.entries(data.headcountByDepartment).map(([name, value]) => ({ name, value }))
26    : [];
27
28  return (
29    <div className="space-y-6">
30      <div>
31        <h2 className="text-2xl font-extrabold text-slate-900 tracking-tight">HR Control Center</h2>
32        <p className="text-slate-500 text-xs mt-0.5">Real-time enterprise organization metrics & operational status</p>
33      </div>
34
35      {/* Stat Cards Grid */}
36      <div className="grid grid-cols-2 lg:grid-cols-4 gap-4">
37        <StatCard title="Total Headcount" value={data?.totalEmployees || 0} trend={` ${data?.activeEmployees || 0} Active`} icon={Users} color="sky" />
38        <StatCard title="New Joiners" value={data?.newJoiners || 0} trend="Past 30 Days" icon={UserPlus} color="emerald" />
39        <StatCard title="On Leave Today" value={data?.employeesOnLeaveToday || 0} icon={Calendar} color="amber" />
40        <StatCard title="Attendance Rate" value={` ${data?.attendanceRateToday || 0}%`} icon={Clock} color="purple" />
41        <StatCard title="Pending Onboarding" value={data?.pendingOnboarding || 0} icon={CheckSquare} color="sky" />
42        <StatCard title="Monthly Payroll" value={` $${(data?.monthlyPayrollCost || 0).toLocaleString()}`} icon={DollarSign} color="emerald" />
43        <StatCard title="Active Exits" value={data?.activeExits || 0} icon={Logout} color="rose" />
44        <StatCard title="Statutory Audit" value="100% Compliant" icon={ShieldCheck} color="purple" />
45      </div>
46
47      {/* Recharts Analytics Charts */}
48      <div className="grid grid-cols-1 lg:grid-cols-3 gap-6">
49        {/* Payroll Trend Area Chart */}
50        <div className="lg:col-span-2 bg-white rounded-2xl p-6 border border-slate-200 shadow-sm space-y-4">
51          <h3 className="text-base font-bold text-slate-800">Payroll Cost Trend ($)</h3>
52          <div className="h-64 w-full">
53            <ResponsiveContainer width="100%" height="100%">
54              <AreaChart data={data?.payrollCostTrend || []}>
55                <XAxis dataKey="month" stroke="#94a3b8" fontSize={12} />
56                <YAxis stroke="#94a3b8" fontSize={12} />
57                <Tooltip />
58                <Area type="monotone" dataKey="cost" stroke="#00286c" fill="#00286c" fillOpacity={0.15} strokeWidth={3} />
59              </AreaChart>
60            </ResponsiveContainer>
61          </div>
62        </div>
63
64        {/* Headcount by Department Pie Chart */}
65        <div className="bg-white rounded-2xl p-6 border border-slate-200 shadow-sm space-y-4">
66          <h3 className="text-base font-bold text-slate-800">Headcount by Department</h3>
67          <div className="h-64 w-full flex items-center justify-center">
68            <ResponsiveContainer width="100%" height="100%">
69              <PieChart>
70                <Pie data={deptData} dataKey="value" nameKey="name" cx="50%" cy="50%" outerRadius={80} label>
71                  {deptData.map((entry, index) => (
72                    <Cell key={`cell-${index}`} fill={COLORS[index % COLORS.length]} />
73                  ))}
74                </Pie>
75              </PieChart>
76            </ResponsiveContainer>
77          </div>
78        </div>
79      </div>
80    </div>
81  );
82 }
83
```

Main.jsx

```
employee-onboarding-system > frontend > src > main.jsx
1  import React from 'react'
2  import ReactDOM from 'react-dom/client'
3  import { Provider } from 'react-redux'
4  import { store } from '../redux/store'
5  import App from './App'
6  import './index.css'
7
8  ReactDOM.createRoot(document.getElementById('root')).render(
9    <React.StrictMode>
10     <Provider store={store}>
11       <App />
12     </Provider>
13   </React.StrictMode>,
14 )
15
```

index.html

```
employee-onboarding-system > frontend > index.html > ...
1  <!doctype html>
2  <html lang="en">
3    <head>
4      <meta charset="UTF-8" />
5      <link rel="icon" type="image/svg+xml" href="/vite.svg" />
6      <meta name="viewport" content="width=device-width, initial-scale=1.0" />
7      <title>Employee Onboarding & HR Management System</title>
8      <link rel="preconnect" href="https://fonts.googleapis.com">
9      <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
10     <link href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700&display=swap" rel="stylesheet">
11   </head>
12   <body class="bg-slate-50 text-slate-900 font-sans antialiased">
13     <div id="root"></div>
14     <script type="module" src="/src/main.jsx"></script>
15   </body>
16 </html>
17
```

APPENDIX II SCREENSHOTS

User Login & Authentication Portal

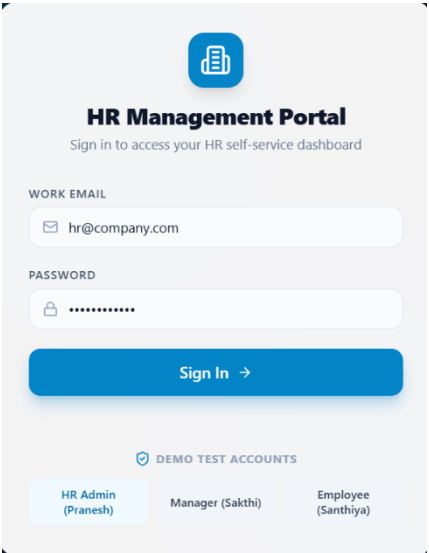


Fig A.2.1: User Login & Authentication Portal

HR Dashboard

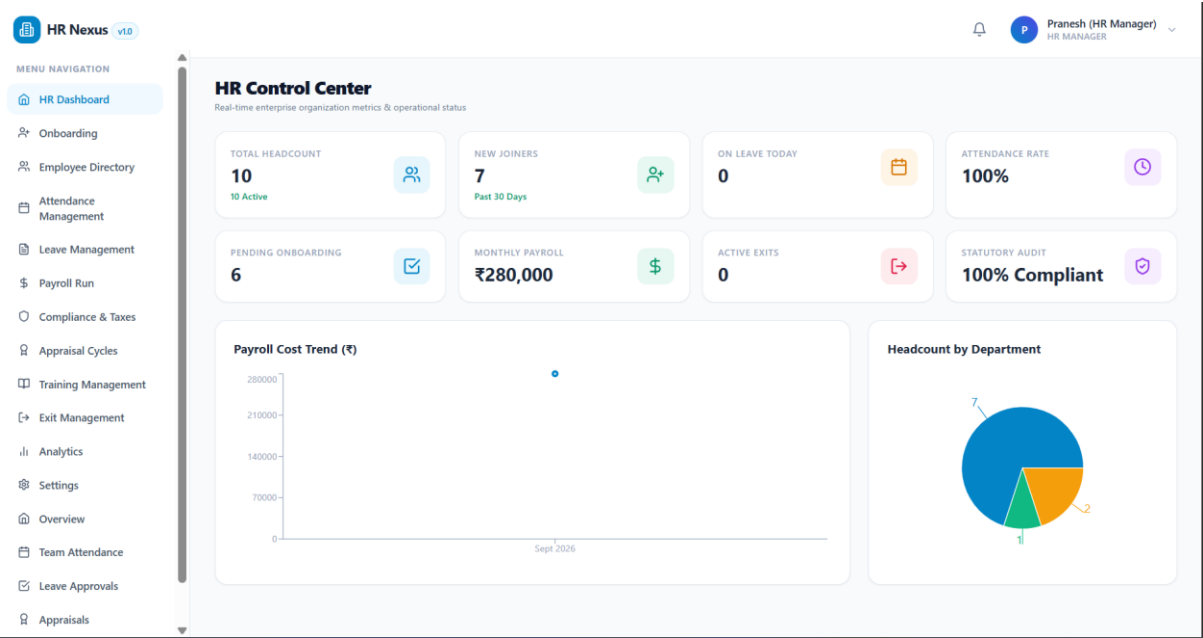


Fig. A.2.2: HR Dashboard

Employee Onboarding

Pre-Joining Onboarding
Generate digital offer letters, track e-sign, BGV, and Day 1 onboarding

Search candidates...

Candidate	Email	Role	Offered CTC	e-Sign	BGV Status	Status	Action
Dhanesh	dhanesh@company.com	Fresher (Engineering)	₹800,000	Mark Signed	Mark BGV	OFFER_GENERATED	Day 1 Onboard
SK	praneshthirumalaivrt@gmail.com	IT Cousantant (Marketing)	₹5,000,000	Mark Signed	Mark BGV	ONBOARDED	Onboarded
vbm	praneshthirumalaivrt@gmail.com	IT (Finance)	₹800,000	Mark Signed	Mark BGV	OFFER_GENERATED	Day 1 Onboard
Dhanesh	Dhanesh@gmail.com	Junior Enginner (Marketing)	₹500,000	Mark Signed	Mark BGV	ONBOARDED	Onboarded
Murgavel	murga@gmail	Enginner (Engineering)	₹500,000	Mark Signed	Mark BGV	ONBOARDED	Onboarded
John Doe	johndoe@example.com	Frontend Developer (Engineering)	₹undefined	Mark Signed	Mark BGV	ONBOARDED	Onboarded
John Doe	johndoe@example.com	Frontend Developer (Engineering)	₹undefined	Mark Signed	Mark BGV	OFFER_ACCEPTED	Day 1 Onboard
Raj	raj@company.com	Freshers (Engineering)	₹800,000	Mark Signed	Passed	BGV_COMPLETED	Day 1 Onboard
Alex	Alex@gmail.com	Enginner (Human Resources)	₹800,000	Mark Signed	Passed	BGV_COMPLETED	Day 1 Onboard
Iyal	iyal@company.com	Fresher (Engineering)	₹800,000	Mark Signed	Mark BGV	OFFER_GENERATED	Day 1 Onboard

Showing 1 to 10 of 10 entries

Fig. A.2.3:Employee Onboarding

A.2.4 EMPLOYEE DATABASE

```
1 • SELECT * FROM employee_onboarding_hr_management.users;
```

id	active	created_at	email	password
1	1	2026-09-06 11:18:35.155845	Iyal@company.com	\$2a\$12\$bKLLQmubUlyE5xIS36aIsu91f0rKFKZa4
1	1	2026-09-05 08:16:15.125820	johndoe@example.com	\$2a\$12\$wRTwNnXYYfEcXyOQcLFQeZ/vURBA
1	1	2026-09-05 08:23:18.041158	loki@gmail.com	\$2a\$12\$5Hbft/JJZBg0IsgRjKcdOwKVCy8dXr
1	1	2026-09-06 11:24:09.265824	raj@company.com	\$2a\$12\$efCg3F63NOS/Maox.XMID6X4Aih5WP,
1	1	2026-09-06 07:35:46.240163	alex@company.com	\$2a\$12\$XuJhB1MwRyByvV.gItYn.V8rMNFoYX,

Fig. A.2.4.1 Users Table

```
1 • SELECT * FROM employee_onboarding_hr_management.employees;
```

employee_id	aadhaar_number	account_number	address	bank_name	basic_salary	created_at
1	987654321098	623401987654	ICICI Bank	47500.00	2026-09-03 05:45	
1	123456789012	99996174581	HDFC Bank	25000.00	2026-09-05 08:16	
1	123456789012	50100123456789	HDFC Bank	60000.00	2026-09-03 05:45	

Fig. A.2.4.2 Employees Table

Payroll Management

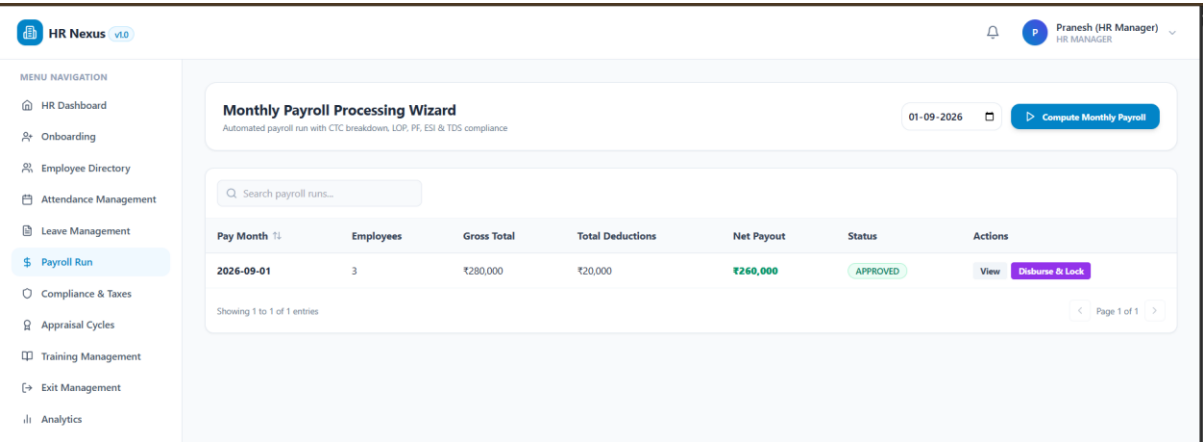


Fig.A.2.4: Payroll Management

A.2.6 Exit & FnF Settlement

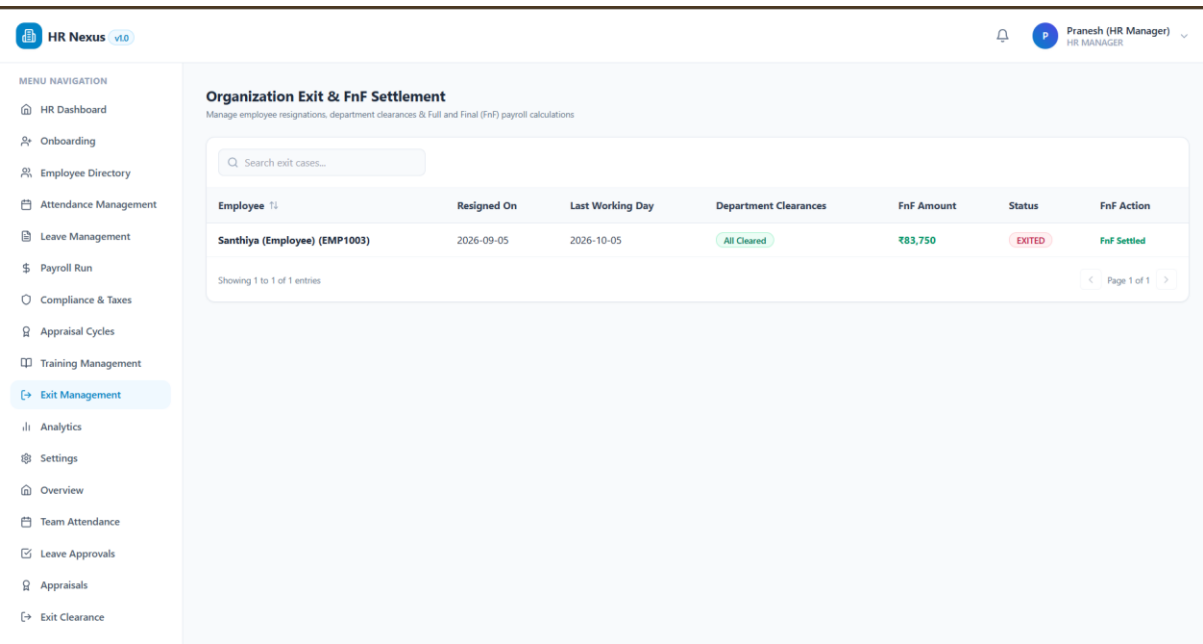


Fig. A.2.6: Exit & FnF Settlement

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